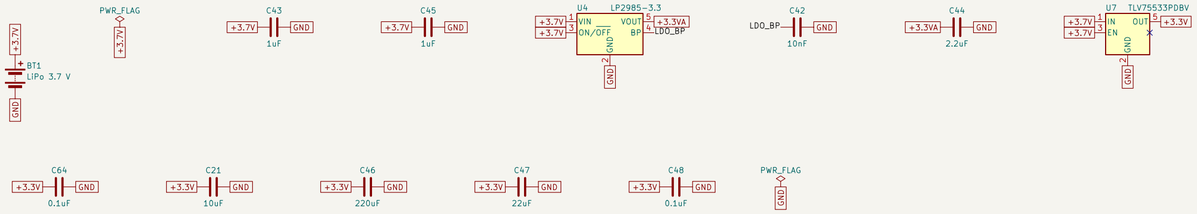
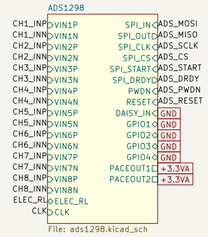
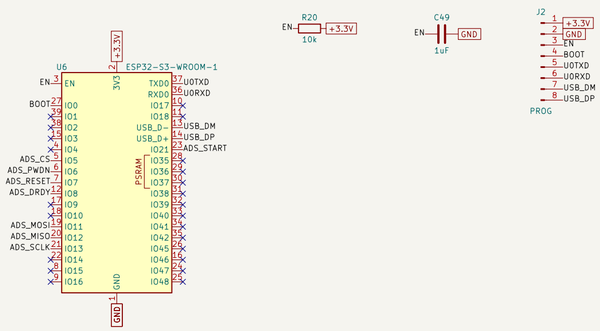
Power: battery, analog LDO U4 -> +3.3VA (ADS1298 AVDD), digital LDO U7 -> +3.3V (DVDD, ESP32-S3)



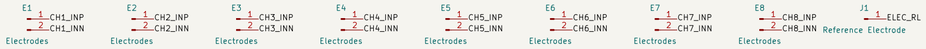
Front end: sheet ads1298.kicad_sch



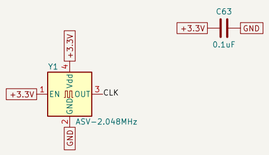
Microcontroller: ESP32-S3-WROOM-1 (SPI to the ADS1298 as on the base design), EN reset RC, programming header



Electrodes: channel pairs E1 to E8 (pin 1 +, pin 2 -), reference (right-leg) electrode J1



ADS1298 master clock: 2.048 MHz oscillator



Sections from the client's base design (open-semg-8), with the supply fixes of the 28 Oct audit: U4 feeds only +3.3VA, U7 only +3.3V, both always on; one ground.
Added: J2 programming header; Y1, the 2.048 MHz clock the ADS1298 core expects (CLKSEL low). SPI as on the base: CS IO5, PWDN IO6, RESET IO7, DRDY IO8, MOSI IO11, MISO IO12, SCLK IO13, START IO21.
Boot: hold J2 pin 4 (IO0) low at reset for download mode. The front end was simulated by the design loop; the board-level sections were checked by netlist and ERC only.

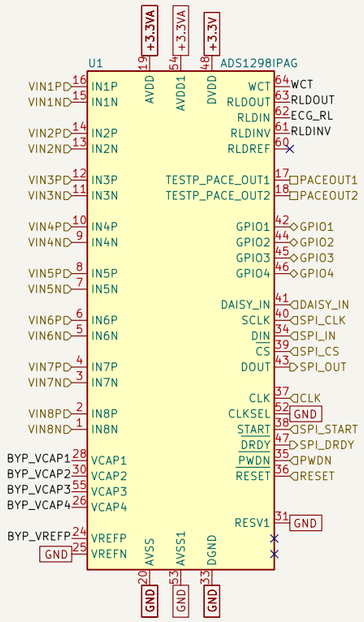
TI terms: ADS1298 only · DigiKey BOM 2026-11-02
Front end: ambrflow design loop, engine e735a0e, vref-0066
Power, MCU, connectors: client's open-semg-8 (audit fixes 28 Oct)

Sheet: /
File: semg8_B_ADS1298.kicad_sch

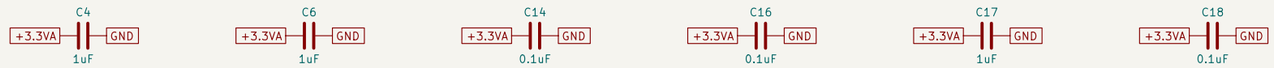
Title: sEMG-8 board B: ADS1298 front end

Size: A2	Date: 2026-11-02	**Rev: A**
KiCad E.D.A. 10.0.6		Id: 1/2

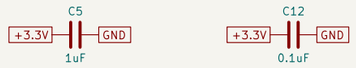
ADS1298, 8-channel 24-bit biopotential AFE (TQFP-64)



Analog supply bypass (AVDD = +3.3VA)



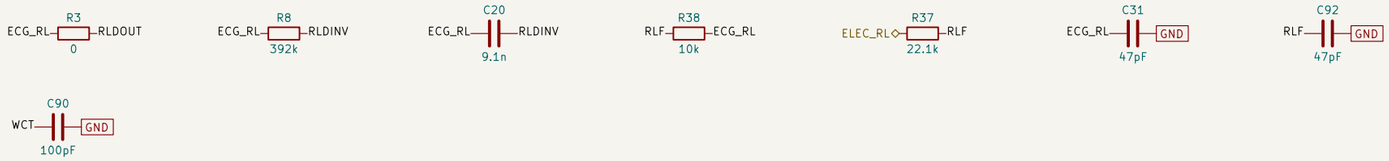
Digital supply bypass (DVDD = +3.3V)



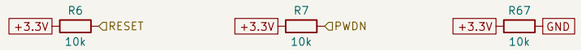
Reference (VREFP) and internal regulator capacitors (VCAP1 to VCAP4)



Right-leg drive and Wilson central terminal



Control pull-ups and DVDD load



TI ADS1298ECG-FE core: AVDD = +3.3VA (LP2985), DVDD = +3.3V (TLV75533), internal 2.4 V reference, right-leg drive out on ELEC_RL.
Ports: the electrode inputs, ELEC_RL (reference electrode J1), CLK (2.048 MHz, Y1), SPI and control to the ESP32-S3; DAISY_IN and GPIO1 to 4 tied low and TESTP/TESTN_PACE_OUT tied to AVDD on the root sheet (the datasheet's ties for unused pins).
RLDREF is open as on TI's sheet: set CONFIG3.RLDREF_INT = 1. C20: TI's 0.01 uF, sourced as 9.1 nF C0G and re-simulated (passes).

Footprints and MPNs: DigiKey BOM 2026-11-02
From vref-0066: TI ADS1298ECG-FE, SBAU171D sheet 2 (TI terms)
ambrflow design loop, engine e735a0e; spec semg8-ads1298 f30ae840439c

Sheet: /ADS1298/
File: ads1298.kicad_sch

Title: sEMG-8 front end: ADS1298 core

Size: A3	Date: 2026-11-02	**Rev: A**
KiCad E.D.A. 10.0.6		Id: 2/2